Dense Dilated Convolutions' Merging Network for Land Cover Classification

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***Abstract*— Land cover classification of remote sensing images is a challenging task due to limited amounts of annotated data, highly imbalanced classes, frequent incorrect pixel-level annotations, and an inherent complexity in the semantic segmentation task. In this article, we propose a novel architecture called the dense dilated convolutions' merging network (DDCM-Net) to address this task. The proposed DDCM-Net consists of dense dilated image convolutions merged with varying dilation rates. This effectively utilizes rich combinations of dilated convolutions that enlarge the network's receptive fields with fewer parameters and features compared with the state-of-the-art approaches in the remote sensing domain. Importantly, DDCM-Net obtains fused local- and global-context information, in effect incorporating surrounding discriminative capability for multiscale and complex-shaped objects with similar color and textures in very high-resolution aerial imagery. We demonstrate the effectiveness, robustness, and flexibility of the proposed DDCM-Net on the publicly available ISPRS Potsdam and Vaihingen data sets, as well as the DeepGlobe land cover data set. Our single model, trained on three-band Potsdam and Vaihingen data sets, achieves better accuracy in terms of both mean intersection over union (mIoU) and F1-score compared with other published models trained with more than three-band data. We further validate our model on the DeepGlobe data set, achieving state-of-the-art result 56.2% mIoU with much fewer parameters and at a lower computational cost compared with related recent work.**

***Index Terms*— Deep learning, land cover classification, semantic segmentation, very high-resolution (VHR) optical imagery.**

I. Introduction

AUTOMATIC semantic classification of land cover in remote sensing data is of great importance for sustainable development, autonomous agriculture, and urban planning. Due to the progress achieved in the deep learning and computer vision community on natural images, most deep learning architectures [1]–[6] for semantic segmentation can also be used for land cover classification tasks in the remote sensing domain. Semantic segmentation refers to the assignment of a semantic category to every pixel in the images, which, in this work, consists of very high-resolution (VHR) aerial images. Currently, the state-of-the-art end-to-end semantic segmentation models are mostly inspired by the idea of fully convolutional networks (FCNs), which generally consists of an encoder–decoder architecture [1]. All layers in the encoder and decoder modules are based on convolutional neural networks (CNN). However, to achieve higher performance, FCN-based end-to-end methods normally rely on deep and wide multiscale CNN architectures that typically require a large number of trainable parameters and computation resources. In addition, there is also a lot of redundancy in deep CNNs that often result in vanishing gradients in backward propagation, diminishing feature reuse in forward propagation, and long training time [7].

In general, VHR remote sensing images contain diverse objects and intricate variations in their aspect ratio and color textures (e.g. roads, roofs, shadows of buildings, low plants, and branches of trees [8]). Moreover, many remote sensing images are completely consisting of "stuff" classes (amorphous regions, such as forest, agricultural areas, and water). This brings challenges for semantic mapping in remote sensing images. Fig. 1 shows illustrative examples of land cover classification for remote sensing images. Thus, richer and multiscale global contextual representations play a key role in land cover mapping for VHR aerial images.

In this article, we propose a novel network architecture, called the dense dilated convolutions' merging network (DDCM-Net), which utilizes multiple dilated convolutions with various dilation rates. The proposed network learns with densely linked dilated convolutions and outputs a fusion of all intermediate features without losing resolutions during the extraction of multiscale features. This significantly reduces the computational redundancies and costs. Our experiments demonstrate that the network achieves robust and accurate results on the representative ISPRS 2-D semantic labeling data sets [8]. Motivated by the recent success of depthwise separable convolutions [10], we also explore grouped convolutions [11] with strided operations adapted into our DDCM-Net, which is shown to further improve net speed and accuracy. We finally demonstrate the effectiveness of the

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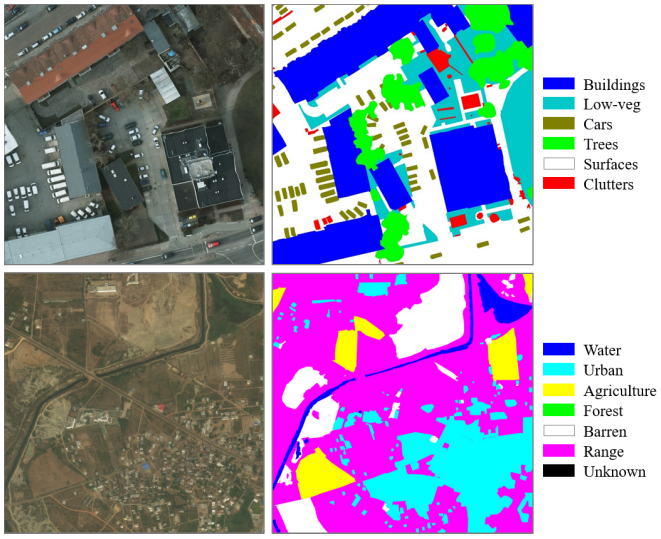


Fig. 1. (Right) Examples of land cover labels and (Left) Corresponding remote sensing images from two different data sets (ISPRS Potsdam [8] and DeepGlobel [9]) separately. Semantic label colors are shown beside the ground truths.

adapted DDCM-Net on the DeepGlobe land cover challenge data set [9]. In summary, our contributions are as follows.

1) We propose a new computationally lightweight and scalable architecture based on dilated convolutions [12] that can be used as a simple, yet effective, encoder or decoder module for semantic segmentation tasks.
2) In the proposed network, one can arbitrarily control the depths, widths, groups, and strides of the modules with various dilation rates in order to address different problems.
3) Our proposed end-to-end model outperforms or achieves competitive performance on different representative remote sensing data sets compared with other published related methods.

A preliminary version of this article appeared in [13]. Here, we extend our work by: 1) extending the methodology with two variants where group and strided convolutions are exploited to further boost the model's flexibility; 2) expanding the experiment section by including more public data sets, providing more detailed training details, and presenting additional result comparisons; 3) providing in-depth discussions in terms of model's dilation and density policies, generalization, as well as a more detailed analysis of computation complexity; and 4) providing a more thorough review of related work.

This article is structured as follows. Section II provides an overview of the related work. Section III introduces the data sets used in our work. In Section IV, we present the methodology in detail. Experimental procedure and evaluation of the proposed method are performed in Section V. Section VI provides discussion of our results. Finally, in Section VII, we draw conclusions.

II. Background

Deep learning and CNNs have been revolutionary for computer vision and image classification [14], [15] in particular. Even though segmentation can be viewed as a pixel-to-pixel classification problem, most modern CNN models for semantic segmentation are inspired by FCNs [1]. The FCN was the first CNN model without any fully connected layers that were trained in an end-to-end manner directly classifying each pixel to its corresponding label. However, vanilla FCNs generally cause loss of spatial information due to the presence of pooling layers that reduce the resolutions of feature maps by sacrificing the positional information of objects. In order to alleviate this issue, U-Net [2] extends the FCN by introducing skip connections between the encoder and decoder modules. In the decoder module, the spatial information is gradually recovered by fusing skipped connections with upsampling layers or deconvolution layers. Since then, the encoder–decoder architecture has been widely extended in recent works, including SegNet [16], GCN [6], PSPNet [4], DUC [5], and DeepLabV3+ [10]. In general, these architectures differ from each other in how they capture rich and global contextual information at multiple scales. For instance, PSPNet [4] introduces a pyramid pooling module to aggregate the context by applying large kernel pooling layers, while DeepLabV3+ [10] utilizes several parallel atrous convolutions with different rates (called Atrous Spatial Pyramid Pooling). Similarly, Schuster *et al.* [17] presented a unified descriptor network for dense matching tasks, so-called SDC-stacked dilated convolution, which combines parallel dilated convolutions with different dilation rates of ($[1, 2, 3, 4]$). Instead of the parallel combination methods, a cascading structure of dilated convolution layers was first presented in [12] with exponentially increasing rates of dilation that achieved state-of-the-art results on natural image segmentation benchmarks in that year. Strubell *et al.* [18] have also proposed a sequential structure of iterating dilated convolutions that demonstrated higher accuracy with impressive speed improvements in contrast to the previously best performing model BI-LSTM-CRF [19], for the sequence labeling tasks when processing entire documents at a time.

In contrast, our novel architecture has three major differences. First, we sequentially stack the output of each layer with its input features before feeding it to the next layer in order to alleviate context information loss. Second, our final output is computed on all features generated by intermediate layers, which can effectively aggregate the fused receptive field of each layer and maximally utilize multiscale context information. Third, our method is much more flexible and extendable with the group and strided convolutions to address different domain problems.

Our applied focus in this article is land cover classification based on remote sensing. Lately, the FCNs and encoder–decoder architectures have been widely adopted and applied to the ISPRS [8] Semantic Labeling Contest [20]–[29] and the DeepGlobe CVPR-2018 [9] challenge of automatic classification of land cover types [30]–[36]. Paisitkriangkrai *et al.* [20] proposed a scheme for high-resolution land cover classification using a combination of a patch-based CNN and a random forest classification that is trained on handcrafted features. Furthermore, Sherrah [21] applied FCNs to semantic labeling of aerial imagery and illustrated that higher

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accuracy can be achieved than with more traditional patch-based approaches. In addition, deep learning has also been exploited for multimodal data processing in remote sensing. For instance, Audebert *et al.* [25] proposed a multiscale SegNet approach (so-called FuseNet) to leverage both a large spatial context and the high-resolution data, while early and late fusion strategies of multimodality data are also exploited. However, such fusion techniques require all modalities to be available to the classification during both training and testing. Kampffmeyer *et al.* [28], therefore, presented a novel CNN architecture based on so-called hallucination networks for urban land cover classification, which is able to replace missing data modalities in the test phase. This enables fusion capabilities even when data modalities are missing in testing.

Recently, Mou and Zhu [27] proposed a recurrent network in FCN (RiFCN) trying to better fuse multilevel features with boundary-aware features to achieve fine-grained inferences. Similarly, the stacked U-Nets architecture is proposed in [34] for ground material segmentation in remote sensing imagery, which merges high-resolution details and the long-distance context information captured at low-resolution to generate segmentation maps. Furthermore, Kuo *et al.* [35] introduced an aggregation decoder in combination with DeepLabV3 architecture to fuse different-level features progressively from the encoder for final prediction, while Tian *et al.* [36] proposed a dense fusion classmate network (DFCNet) that tried to fuse auxiliary training data as "classmate" to capture supplementary features for land cover classification. One of the main ideas behind all the architectures is to take into account the multilevel context to improve the prediction of the segmentation. Even though these state-of-the-art designs could alleviate the loss of global contextual information, they are often computationally expensive with a lot of redundancy in order to capture dense and multiscale contextual features [7]. In the following, we demonstrate that the DDCM-Net achieves competitive results or outperforms for land cover classification on benchmark data sets at a lower computational cost compared with related recent work.

III. Benchmark Data Sets

In this article, we focus on two publicly used databases, namely, the ISPRS 2-D semantic labeling contest data sets [8] and the DeepGlobe land cover challenge data set [9]. The ISPRS data sets are comprised of aerial images over two cities in Germany, i.e., Potsdam[1] and Vaihingen,[2] which have been labeled with six of the most common land cover classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter. The DeepGlobe land cover data set consists of satellite data collected from the DigitalGlobe Vivid+ data set [9] and focuses on rural areas. This includes seven types of land covers: urban (man-made, built-up areas with human artifacts), agriculture (farms, cropland, orchards, vineyards, ornamental horticultural areas, and so on), rangeland (any nonforest, nonfarm, green land, and grass), forest (any land with at least 20% tree crown density plus clear cuts), water (rivers, oceans, lakes, wetland, and ponds), barren (mountain, rock, desert, beach, and land with no vegetation), and unknown (clouds and others). Each data set provides online leaderboards and reports test metrics measured on hold-out test images.

A. Potsdam

The Potsdam data set consists of 38 tiles of size 6000 × 6000 pixels with a ground resolution of 5 cm; 14 of these are used as hold-out test images. Tiles consist of red–green–blue–infrared (RGB-IR) four-channel images. While both the digital surface model (DSM) and normalized DSM (nDSM) data are also included in the data set, we only focus on the three-channel RGB images in this work.

B. Vaihingen

The Vaihingen data set contains 33 tiles of varying sizes (on average approximately 2100 × 2100 pixels) with a ground resolution of 9 cm, of which 17 are used as hold-out test images. Tiles are composed of infrared–red–green (IRRG) three-channel images. Though DSMs and nDSMs data are also available for all images in the data set, we only use IRRG data in this article.

C. DeepGlobe

DeepGlobe Land Cover data contains 1146 RGB images of size 2448 × 2448 pixels with a ground resolution of 50 cm; 803 of these images have a publicly available ground truth and are used as the training set, while the remaining images are split into a hold-out validation and test set consisting of 171 and 172 images, respectively. Due to the variety of land cover types and density of annotations [9], this data set is more challenging than the two above-mentioned data sets.

IV. Methods

We first briefly revisit the concept of dilated convolutions. We then present our proposed DDCM architecture based on such dilated convolutions. Furthermore, we provide detailed information regarding the procedure for training the network.

A. Dilated Convolutions

Dilated convolutions [12] have been demonstrated to improve performance in many classification and segmentation tasks [10], [11], [37], [38]. One key advantage is that they allow us to flexibly adjust the filter's receptive field to capture multiscale information without resorting to downscaling and upscaling operations. A 2-D dilated convolution operator can be defined as

$$g(x_\ell) = \sum_{c \in C_\ell} \theta^c_{k,r} * x^c_\ell \tag{1}$$

where $*$ denotes a convolution operator, and $g : \mathbb{R}^{H_\ell \times W_\ell \times C_\ell} \rightarrow \mathbb{R}^{H_{\ell+1} \times W_{\ell+1}}$ convolves the input feature map $x_\ell \in \mathbb{R}^{H_\ell \times W_\ell \times C_\ell}$. A dilated convolution $\theta_{k,r}$ with a filter size k and dilation rate

[1] http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-potsdam.html

[2] http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

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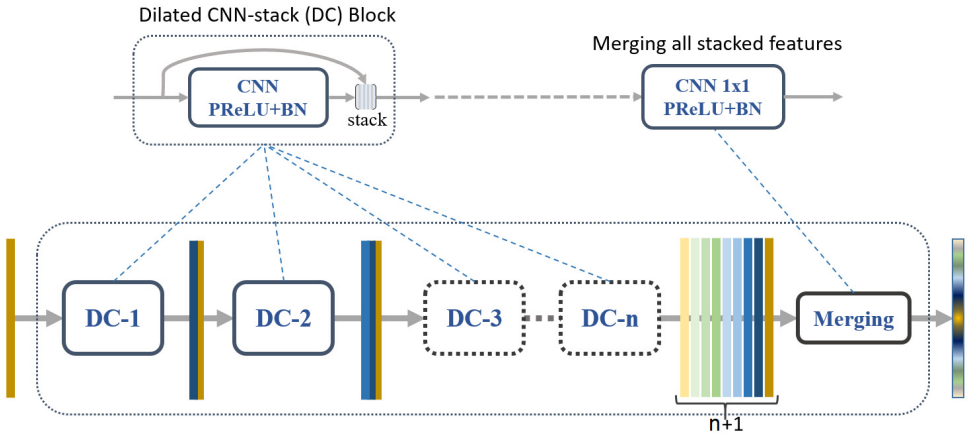


Fig. 2. Example of the DDCM architecture composed of n $\{1, 2, 3, \ldots, n\}$ DC blocks with various dilation rates.

$r \in \mathbb{Z}^+$ is only nonzero for a multiple of r pixels from the center. In a dilated convolution, a kernel size k is effectively enlarged to $k + (k-1)(r-1)$ with the dilation factor r. As a special case, a dilated convolution with dilation rate $r = 1$ corresponds to a standard convolution.

B. Dense Dilated Convolutions' Merging Module

The dense dilated convolutions' merging module (DDCM-Module) consists of a number of dilated CNN-stack (DC) blocks with a merging module as output, as shown in Fig. 2. A basic DC block is composed of a dilated convolution followed by PReLU [39] nonlinear activation and batch normalization (BN) [40]. It then stacks the output with its input and feeds the stacked data to the next layer. The final network output is computed by a merging layer composed of 1×1 filters with PReLU and BN in order to efficiently combine all stacked features generated by intermediate DC blocks. In practice, densely connected DC blocks, typically configured with linearly or exponentially increasing dilation factors, enable DDCM networks to have very large receptive fields with just a few layers. Please note that we apply zero-padding to the input of every DC block in order to keep the resolution of its output equal to the resolution of the input.

Fig. 3 shows a simple 1-D example of the DDCM module composed of three DC blocks (kernel size equal to 3) with dilation rates of $[1, 2, 4]$. In the DC-1 layer, x_1 is produced from x_0 by a one-dilated convolution, where each element of x_1 has a receptive field of 3. In the DC-2 layer, x_2 is produced from $[x_1, x_0]$ by a two-dilated convolution. Note that the receptive field for the elements of x_2 is $[7, 5]$. Similarly, in the DC-3 layer, x_3 is produced from $[x_2, x_1, x_0]$ by a four-dilated convolution with fused receptive fields of $[15, 11, 9]$. The final output x^* is, thus, produced from $[x_3, x_2, x_1, x_0]$ by the so-called merging layer, which fuses multiscale context information by aggregating various receptive fields of $[15, 11, 9, 7, 5, 3, 1]$. It is easy to see that the number of parameters associated with each DC layer grows linearly, while the fused receptive field size is nearly exponentially increasing.

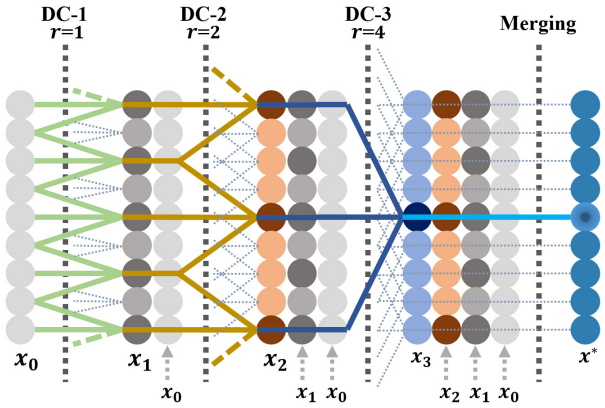


Fig. 3. Simple 1-D example of the DDCM module composed of three DC blocks (kernel size = 3) with dilation rates of 1, 2, and 4. Here, we can see that x_1 is produced from x_0 by a one-dilated convolution with a receptive field of 3. x_2 is produced from $[x_1, x_0]$ by a two-dilated convolution with fused receptive fields of $[7, 5]$. x_3 is produced from $[x_2, x_1, x_0]$ by a four-dilated convolution with fused receptive fields of $[15, 11, 9]$. The final output x^* is produced from $[x_3, x_2, x_1, x_0]$ by the so-called merging layer, which fuses multiscale context information by aggregating various receptive fields of $[15, 11, 9, 7, 5, 3, 1]$.

C. Variants of the DDCM Module

1) Grouped Convolutions: Inspired by ResNeXt [41], grouped convolutions [14] are also exploited in the DC blocks in order to further reduce the depth and parameter size of DDCM-net, especially when the DDCM modules are used as the decoders of high-level features. ResNeXt has demonstrated that increasing cardinality (group number) is a more effective way of gaining accuracy than going deeper or wider [41]. We, therefore, introduce a variant of DDCM, i.e. DDCM($g = 2$), where "$g = 2$" denotes the fact that grouped convolutions with two groups are used in the DC blocks.

2) Strided Convolutions: To further reduce the computational cost when increasing dilation rates, we can apply a dilated convolution with a stride of greater than one pixel, which samples only every s pixels in each direction in the output. Here, s denotes the stride of this dilated convolution. This is similar to a two-step approach with unit stride convolution followed by downsampling but reduces the computational cost.

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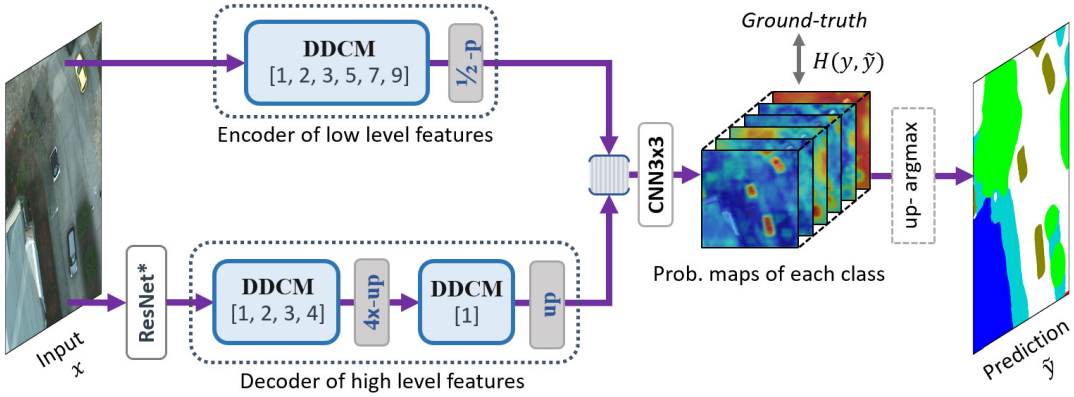


Fig. 4. End-to-end pipeline of DDCM-Net for semantic mapping of VHR Potsdam images. The encoder of low-level features encodes multiscale contextual information from the initial input images by a DDCM module (output three-channel) using 3×3 kernels with six different dilation rates [1, 2, 3, 5, 7, 9]. The decoder of high-level features decodes highly abstract representations learned from a ResNet-based backbone (output 1024-channel) by two DDCM modules with rates [1, 2, 3, 4] (output 36-channel) and [1] (output 18-channel) separately. The transformed low- and high-level feature maps by DDCMs are then fused together to infer pixelwise class probabilities. Here, "p" and "up" denote pooling and upsampling, respectively.

When using a strided dilated convolution in a DC block, we, therefore, need to apply bilinear upsampling to scale the output to the same resolution as the input before concatenation. There are three variants of the DDCM-Module evaluated in this article, i.e. DDCM($s = 2$), DDCM($s = 3$), and DDCM($s = r + 1$), with different striding strategies: a stride of 2, a stride of 3, and a dynamic stride of $(r + 1)$ separately, where r denotes the corresponding dilation rate.

D. DDCM Network

DDCM modules define building blocks from which a more complex network can be built. This is illustrated in Fig. 4 showing the end-to-end pipeline of the DDCM-Net combined with a pretrained model for land cover classification. Compared with other encoder–decoder architectures, our proposed DDCM-Net only fuses low-level features one time before the final prediction CNN layers, instead of aggregating multiscale features captured at many different encoder layers [2], [6], [27], [31]–[36]. This makes our model simple and neat, yet effective with lower computational cost. In particular, this model is easy to adapt by adjusting the density (number of the output feature maps) and dilation strategy of the encoder and/or decoder features to tackle different tasks, depending on different domains.

In this work, we only utilize the first three bottleneck layers of pretrained ResNet-based [42] backbones (both ResNet50 [42] and SE-ResNeXt50 [43]) and remove the last bottleneck layer and the fully connected layers to reduce the number of parameters to train. Furthermore, due to the larger complexity and a variety of the DeepGlobe data set compared with the ISPRS data, we utilize a DDCM($s = 2$) module configured with larger dilation growing rates [1, 2, 4, 8, 16, 32] as the low-level encoder and two DDCM($g = 2, s = 2$) modules configured by [1, 2, 4] and [1] as the high-level decoder. This configuration results in feature maps of size 64 and 32 channels, rather than 36 and 18 channels for the model on ISPRS data. We also choose SE-ResNeXt50 as the backbone, instead of ResNet50.

V. Experiments and Results

In this section, we investigate the proposed network on the Potsdam (Section V-C), Vaihingen (Section V-C), and DeepGlobe (Section V-D) data sets and report both qualitative and quantitative results of multiclass land cover classification.

A. Training Details

According to best practices, we train using Adam [44] with AMSGrad [45] as the optimizer with weight decay 2×10^{-5} applied to all learnable parameters except biases and batch-norm parameters and polynomial learning rate (LR) decay $(1-(\text{cur_iter}/\text{max_iter}))^{0.9}$ with the maximum iterations of 10^8. We also set $2 \times LR$ to all bias parameters in contrast to weights parameters. We use initial LRs of $(8.5 \times 10^{-5}/\sqrt{2})$ and $(8.5 \times 10^{-4}/\sqrt{2})$ for the ISPRS data and DeepGlobe data, respectively. For the training on ISPRS data (both Potsdam and Vaihingen), we utilized a stepwise LR schedule method that reduces the LR by a factor of 0.85 every 15 epochs based on our training observations and empirical evaluations, while for the training on DeepGlobe data, we utilized multistep LR policy, which reduces the LR by a factor of 0.56 at epochs [4, 8, 16, 24, 32, 96, 128] guided by our empirical results.

We apply a cross-entropy loss function with median frequency balancing (MFB) weights as defined in (2) and (3) [46]

$$W_c = \frac{\text{median}(\{f_c | c \in \mathcal{C}\})}{f_c} \tag{2}$$

$$\text{Loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} {l_c}^{(n)} \log\left({p_c}^{(n)}\right) W_c \tag{3}$$

where W_c is the weight for class c, f_c is the pixel-frequency of class c, ${p_c}^{(n)}$ is the probability of sample belonging to class c, and ${l_c}^{(n)}$ denotes the class label of sample n in class c.

We train and validate the networks for the Potsdam and Vaihingen data sets with randomly sampled 5000 patches of size 256×256 as input and batch size of five. For the experiments on the DeepGlobe data set, we use two downsampled resolutions of 1224×1224 and 816×816 (downscaled

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from 2448 × 2448), and then, we train with 4000 crops (765 × 765) and batch size of four. The training data are sampled uniformly and randomly shuffled for each epoch. We conduct all experiments in this article using PyTorch [47] on a single computer with one NVIDIA 1080Ti GPU.

B. Augmentation and Evaluation Methods

During training on Potsdam and Vaihingen data sets, we randomly flip or mirror images for data augmentation (with probability 0.5), while on the DeepGlobe data, we also augment the crops by randomly shifting (limit 0.0625), scaling (limit 0.1), and rotating them (limit 10). The albumentations library [48] for data augmentation is utilized in this article. Please note that all training images are normalized to [0.0, 1.0] after data augmentation.

We apply test time augmentation (TTA) in terms of flipping and mirroring. For the Potsdam and Vaihingen data sets, we use sliding windows (with 448 × 448 size at a 100-pixel stride) on a test image and stitch the results together by averaging the predictions of the overlapping TTA regions to form the output. While for DeepGlobe data, we first apply TTA on a downsampled (3×) test image (816 × 816) and then upsample all the predictions back to original sizes and average them to get the final output. The performance is measured by both the F1-score [46] and the mean Intersection over Union (IoU) [49]. Please note that the mIoU metric was computed by averaging over the six classes (excluding the "Unknown" class) in the DeepGlobe contest.

C. Potsdam and Vaihingen

For evaluation, the labeled part of the Potsdam data set is split into a training set (19 images), a validation set (two images of 4_10 and 7_10), and a local test set (three images of areas 5_11, 6_9, and 7_11). The Vaihingen data set is similarly divided into training (11 images), validation (two images of areas 7 and 9), and local test set (four images of areas 5, 15, 21, and 30). While the hold-out test sets contain 14 images (areas: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13) and 17 images (areas: 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38) for the Potsdam and Vaihingen data sets, respectively. Table I shows our results on the hold-out test sets and our local test sets of ISPRS Potsdam and Vaihingen separately with a single trained model. The mean F1-score (mF1) and the mean IoU (mIou) are computed as the average measure of all classes except the clutter class.

We also compare our results with other related published work on the ISPRS Potsdam RGB data set and Vaihingen IRRG data set. These results are shown in Tables II and III, respectively. Our single model with overall F1-score (92.3%) on Potsdam RGB data set achieves around 0.5 % higher score compared with the second-best model—FuseNet+OSM [25]. Similarly, our model trained on Vaihingen IRRG images also obtained the best overall performance with 89.8% F1-score that is around 1.1% higher than the score of the second-best model—GSN [26]. It is also worth noting that although our OA is only marginally better (+0.1%) for Vaihingen and even worse (−1.5%) for Potsdam, our model obtained better F1 scores. We, therefore, believe that our proposed method has better capability to handle extremely unbalanced classes. Furthermore, by balancing and modeling the surrounding classes (such as road/surface and buildings) more accurately with our model, the car class will be easier to distinguish and, thus, has better results with increased receptive fields, as shown in Table III.

Fig. 5 shows the qualitative comparisons of the land cover mapping results from our model and the ground truths on the test set. We observe that our model is able to segment both large multiscale objects (such as buildings) and small objects (such as cars) very well with fine-grained boundary recovery without any postprocessing.

D. DeepGlobe

Our DDCM-SER50 model achieves a new state-of-the-art result with 56.2% mIoU on the DeepGlobe land cover classification challenge data set. As shown in Table IV, we compare our DDCM network with other published models [30]–[36] on the hold-out validation set (the public

TABLE I

RESULTS ON THE HOLD-OUT TEST IMAGES OF ISPRS POTSDAM AND VAIHINGEN DATA SETS WITH A SINGLE TRAINED DDCM-R50 MODEL SEPARATELY

F1-score	OA	Surface	Building	Low-veg	Tree	Car	mF1
Potsdam	0.908	0.929	0.969	0.877	0.894	0.949	0.923
	0.931*	0.946*	0.983*	0.865*	0.892*	0.939*	0.925*
Vaihingen	0.904	0.927	0.953	0.833	0.894	0.883	0.898
	0.921*	0.934*	0.973*	0.814*	0.914*	0.909*	0.909*
IoU							**mIoU**
Potsdam	0.908	0.867	0.940	0.781	0.809	0.902	0.860
	0.931*	0.898*	0.966*	0.762*	0.805*	0.885*	0.863*
Vaihingen	0.904	0.863	0.909	0.713	0.808	0.790	0.817
	0.921*	0.876*	0.948*	0.686*	0.842*	0.832*	0.837*

* Results marked with * were measured on our local test images, others were measured on hold-out test sets (14 images and 17 images for the Potsdam and Vaihingen separately).

TABLE II

COMPARISONS BETWEEN OUR METHOD WITH OTHER PUBLISHED METHODS ON THE HOLD-OUT RGB TEST IMAGES OF ISPRS POTSDAM DATA SET

Models	OA	Surface	Building	Low-veg	Tree	Car	mF1
HED+SEG.H-Sc1 [23]	0.851	0.850	0.967	0.842	0.686	0.858	0.846
RiFCN [27]	0.883	0.917	0.930	0.837	0.819	0.937	0.861
RGB+I-ensemble [28]	0.900	0.870	0.936	0.822	0.845	0.892	0.873
Hallucination [28]	0.901	0.873	0.938	0.821	0.848	0.882	0.872
DNN_HCRF [29]	0.884	0.912	0.946	0.851	0.851	0.928	0.898
SegNet RGB [25]	0.897	0.930	0.929	0.850	0.851	0.951	0.902
DST_2 [21]	0.903	0.925	0.964	0.867	0.880	0.947	0.917
FuseNet+OSM [25]	**0.923**	**0.953**	0.959	0.863	0.851	**0.968**	0.918
Ours							
DDCM-R50	0.908 (-1.5%)	0.929 (-2.4%)	**0.969** (+0.5%)	**0.877** (+1.0%)	**0.894** (+1.4%)	0.949 (-1.9%)	**0.923** (+0.5%)
DDCM($s = 2$)	0.908	0.930	0.968	0.876	0.895	**0.952**	**0.924**
DDCM($s = 3$)	0.910	0.932	0.967	0.878	0.895	0.937	0.922
DDCM($s = r + 1$)	0.911	0.933	0.968	0.876	0.894	0.950	0.924

TABLE III

COMPARISONS BETWEEN OUR METHOD WITH OTHER PUBLISHED METHODS ON THE HOLD-OUT IRRG TEST IMAGES OF ISPRS VAIHINGEN DATA SET

Models	OA	Surface	Building	Low-veg	Tree	Car	mF1
UOA [22]	0.876	0.898	0.921	0.804	0.882	0.820	0.865
DNN_HCRF [29]	0.878	0.901	0.932	0.814	0.872	0.720	0.848
ADL_3 [20]	0.880	0.895	0.932	0.823	0.882	0.633	0.833
DST_2 [21]	0.891	0.905	0.937	0.834	0.892	0.726	0.859
ONE_7 [24]	0.898	0.910	0.945	**0.844**	0.899	0.778	0.875
DLR_9 [23]	0.903	0.924	0.952	0.839	0.899	0.812	0.885
GSN [26]	0.903	0.922	0.951	0.837	**0.899**	0.824	0.887
Ours							
DDCM-R50	**0.904** (+0.1%)	**0.927** (+0.3%)	**0.953** (+0.1%)	0.833 (-1.1%)	0.894 (-0.5%)	**0.883** (+5.9%)	**0.898** (+1.1%)
DDCM($s = 2$)	0.901	0.924	0.951	0.826	0.891	**0.890**	0.896
DDCM($s = 3$)	0.901	0.923	0.951	0.828	0.891	0.879	0.894
DDCM($s = r + 1$)	0.901	0.923	0.949	0.829	0.892	0.888	0.896

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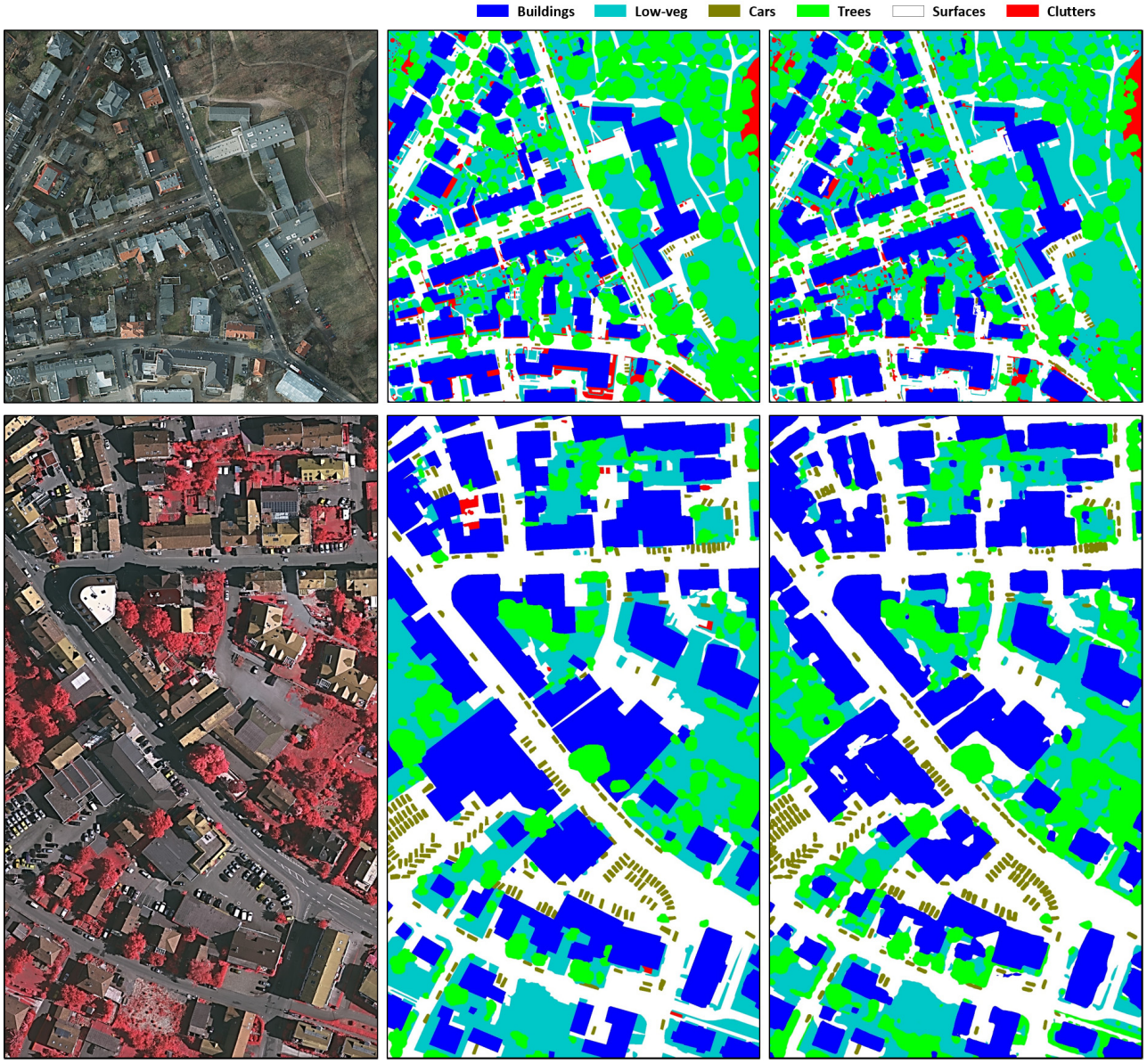


Fig. 5. Mapping results for test images of Potsdam tile-3_14 (top) and Vaihingen tile-27 (bottom). (From left to right) (Left) Input images, and (Middle) ground truths and the predictions of our single DDCM-R50 model.

leaderboard[3] up to the date of May 1, 2019). Our model obtained above 3.5% higher mIoU than the second-best model [35]. Fig. 6 shows some examples of the predictions on the hold-out validation images. We did not apply any postprocessing (e.g., graph-based fine-tuning as utilized in [35]) nor any multiscale prediction fusion methods as applied in [36]. We just make the prediction on the 3× downscaling of the test image and then perform upsampling back to its original resolution.

VI. Discussion

A. Preliminary Analysis

As a baseline, we reimplemented, trained, and evaluated some popular architectures on the local Potsdam test set [49].

[3]https://competitions.codalab.org/competitions/18468#results

TABLE IV

Performance Comparisons on the Hold-Out Validation Set of DeepGlobe Data With Other Published Methods

Models	mIoU	GFLOPs
NU-Net [30]	0.428	-
InceptionV3+Haralick [31]	0.476	-
GCN-based [32]	0.485	-
FPN [33]	0.493	-
Stacked U-Nets [34]	0.507	-
DeepLabv3+ [35]	0.510	-
ClassmateNet [36]	0.519	-
DFCNet [36]	0.526	-
Deep Aggregation Net [35]	0.527	-
Ours		
DDCM-SER50	**0.562**	**4.68**

We compared our methods to them in terms of parameter sizes, computational cost (FLOPs), inference time on both CPU (AMD Ryzen Threadripper 1950X) and GPU (NVIDIA GeForce GTX 1080Ti), and mIoU evaluated on the full

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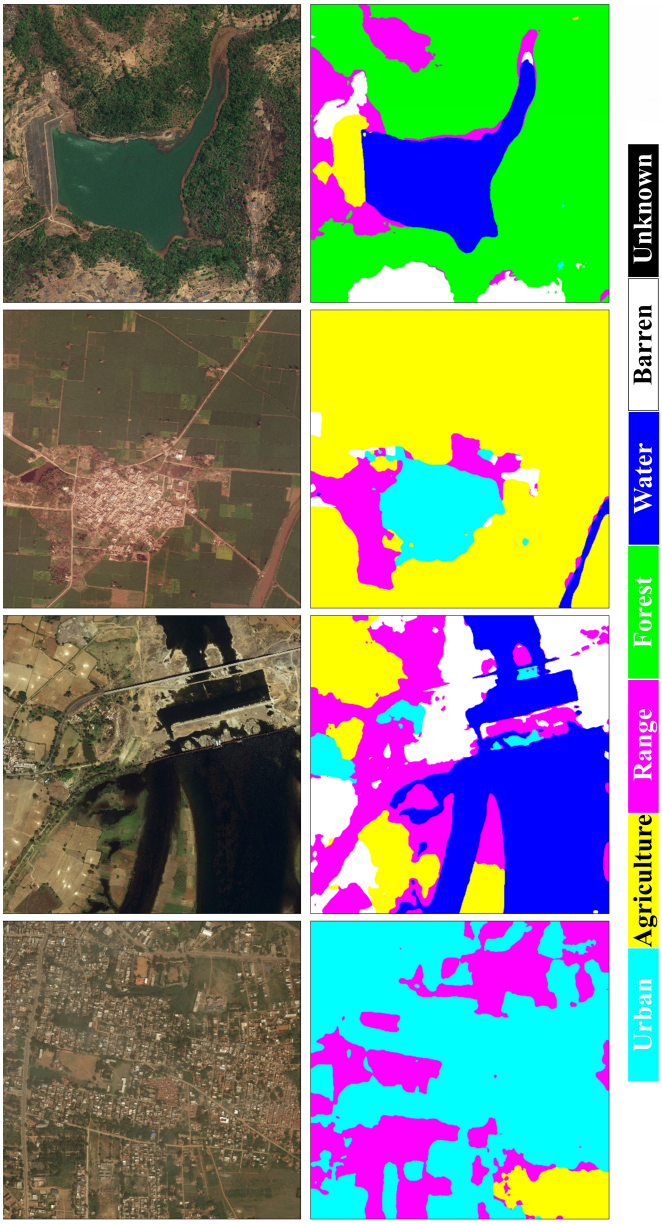


Fig. 6. Mapping results on hold-out validation images of DeepGlobe. (From left to right) Input satellite images and the predictions of our model.

reference ground truths of the data set. Table V details the quantitative results of our DDCM-R50 model against others. Our model consumes about 9× and 13× less FLOPs with 4× and 4.7× fewer parameters and 12× and 24× faster inference speed on CPU but achieves +1.9% and +2.7% higher mIoU than PSPNet [4] and SegNet [3] respectively.

In addition, we also investigated the effectiveness of strided convolutions with the purpose of reducing the computational cost of dilated modules. We observe that with a dynamic stride of $r + 1$, our model has the best speed without loss of the F1-scores in comparison to a stride of 2 or stride-1 model as shown in the final hold-out tests in Table II. Interestingly, we find that both dynamic strided and stride-2 policies could improve the models' accuracy of small objects (i.e., cars). Overall, DDCM($s = 2$)-based mode obtained the

TABLE V

QUANTITATIVE COMPARISON OF PARAMETERS SIZE, FLOPS (MEASURED ON INPUT IMAGE SIZE OF 3 × 256 × 256), INFERENCE TIME ON CPU AND GPU SEPARATELY, AND MIOU ON ISPRS POTSDAM RGB DATA SET

Models	Backbones	Parameters (Million)	FLOPs (Giga)	Inference time (ms - CPU/GPU)	mIoU*
U-Net [2]	VGG16	31.04	15.25	1460 / **6.37**	0.715
FCN8s [1]	VGG16	134.30	73.46	6353 / 20.68	0.728
SegNet [3]	VGG19	39.79	60.88	5757 / 15.47	0.781
GCN [6]	ResNet50	23.84	5.61	593 / 11.93	0.774
PSPNet [4]	ResNet50	46.59	44.40	2881 / 81.08	0.789
DUC [5]	ResNet50	30.59	32.26	2086 / 68.24	0.793
Ours					
DDCM-R50	ResNet50†	**9.99**	**4.86**	**238** / 10.23	**0.808**
DDCM($s = 2$)	ResNet50†	9.99	4.48	159 / 11.39	**0.811**
DDCM($s = 3$)	ResNet50†	9.99	4.43	144 / 11.25	0.798
DDCM($s = r + 1$)	ResNet50†	9.99	4.42	**132** / 11.50	0.810

* mIoU was measured on full reference ground truths of our local test images 5_11, 6_9 and 7_11 in order to fairly compare with our previous work [49].
† Inference time was measured on CPU - AMD Ryzen Threadripper 1950X and GPU - NVIDIA GeForce GTX 1080Ti respectively.

best performance on small objects (cars) with +0.3% and +0.7% higher IoU than standard convolutions on both the Potsdam and Vaihingen test data sets, respectively, as shown in Tables II and III. We, therefore, believe that dilated convolutions with strided operations (i.e., a stride of 2) could not only improve a model's computational efficiency but also capture better contextual representations that further boost the model's capability for detailed object boundary recovery.

B. Flexible Dilation and Density Policies

We used two different dilation settings strategies. For ISPRS data, we configured a DDCM module with linearly growing dilation rates ([1, 2, 3, 5, 7, 9]) as the low-level features encoder, while for the DeepGlobe data set, we build the encoder with exponentially growing dilation rates ([1, 2, 4, 8, 16, 32]) with a stride of two since we see that the DeepGlobe images contain more spatially chaotic objects with lower resolutions, larger scales, and less geometrical attributes than the ISPRS images. We believe DDCM modules with bigger dilation configurations could capture the larger multiscale and global context in this case but require more computational cost. Hence, one has to make some tradeoffs on the dilation policies and the densities based both on the data set and on the budget of computation resources. Similarly, as for the decoders of high-level features, we also follow different strategies in terms of the dilation settings and densities of DDCM modules for the ISPRS and DeepGlobe data, as described in Section IV-D. In particular, we adopt both strided and grouped convolutions together that use stride equal to two and group equal to two in the DDCM modules with the output densities of 64 and 32 channels, respectively. These settings strike the best tradeoff between the speed and accuracy of the DeepGlobe database in our experiments.

We also evaluated the influence of the low-level feature encoder and dilated strategies. We performed two ablation studies by training the following two models: 1) the no-LL-encoder model that was configured without the low-level encoder stream but only used the high-level decoder branch which output 21 channels instead of 18 channels in the standard DDCM-R50 model and 2) the no-dilation model that only uses standard convolutions by fixing the dilation rate

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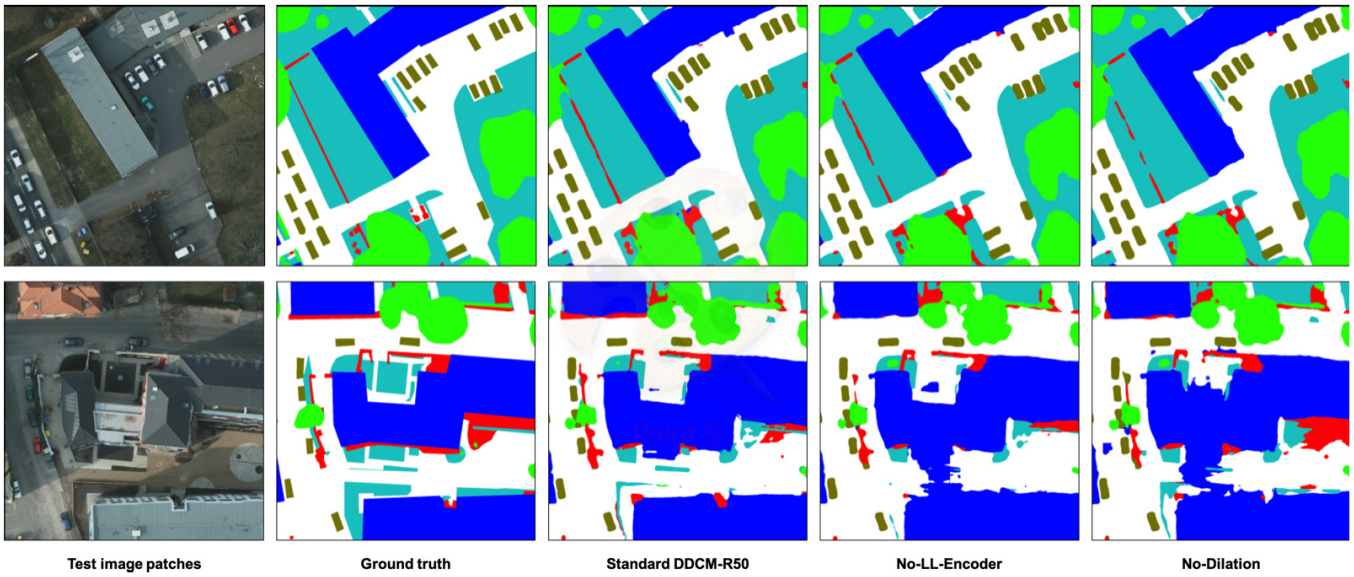


Fig. 7. Mapping results comparison. (From left to right) Input image patches, (left) ground truth, the standard DDCM-R50, the no-low-level-encoder DDCM-R50, and (right) no-dilation DDCM-R50.

TABLE VI

ABLATION STUDIES FOR OUR PROPOSED METHOD ON THE HOLD-OUT RGB TEST IMAGES OF THE ISPRS POTSDAM DATA SET

Models	OA	Surface	Building	Low-veg	Tree	Car	mF1
DDCM-R50	**0.908**	**0.929**	**0.969**	**0.877**	**0.894**	**0.949**	**0.923**
No-LL-Encoder	0.899 (-0.9%)	0.919 (-1.0%)	0.948 (-2.1%)	0.869 (-0.8%)	0.893 (-0.1%)	0.936 (-1.3%)	0.913 (-1.0%)
No-Dilation	0.892 (-1.6%)	0.910 (-1.9%)	0.938 (-3.1%)	0.868 (-0.9%)	0.892 (-0.2%)	0.929 (-2.0%)	0.908 (-1.5%)

* No-LL-Encoder means the model was configured without the low-level encoder stream, while adjusting the output channels of the high level decoder to 21 instead of 18 in the standard DDCM-R50 model. The reason for increasing the number of the high level channels is to counter the loss of features from the low-level encoder stream.
* No-Dilation means that each DDCM module in the DDCM-R50 model only used unit dilation rates for its convolution layers.

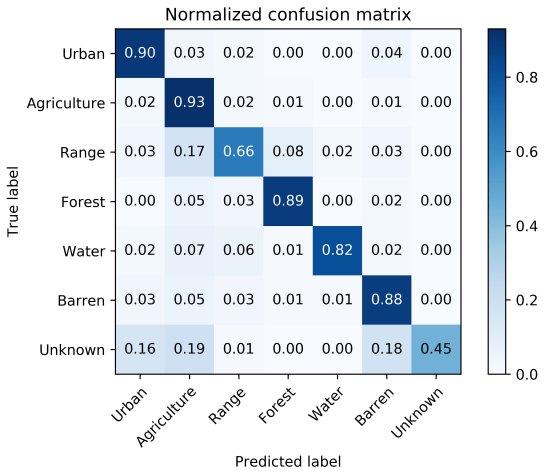
TABLE VII

IOU SCORES OF OUR 5 K-FOLD MODELS ON LOCAL VALIDATION SETS OF DEEPGLOBE DATA SET

K-Fold	Urban	Agriculture	Range	Forest	Water	Barren	mIoU
k0	0.783	**0.901**	**0.488**	0.760	0.605	0.723	0.710
k1	0.735	0.876	0.391	0.772	0.730	**0.739**	0.707
k2	**0.821**	0.873	0.381	0.757	0.832	0.723	0.731
k3	0.796	0.883	0.421	0.789	0.849	0.716	**0.742**
k4	0.724	0.830	0.431	**0.795**	**0.857**	0.654	0.715
Avg.	**0.772**	**0.872**	**0.422**	**0.775**	**0.775**	**0.711**	**0.721**

TABLE VIII

NORMALIZED CONFUSION MATRIX ON A LOCAL VALIDATION SET OF DEEPGLOBE DATA



$r = 1$ for each of the convolutional layers in the DDCM-R50 model. Table VI shows the final test results on the Potsdam data set. The performance, in terms of mF1, dropped off overall 1.0% and 1.5% with no-LL-encoder and no-dilation, respectively. Performance is particularly decreasing for the building (−2.1% and −3.1%) and car (−1.3% and −2.0%) classes. Their findings are also validated by the qualitative visualization of the results that are shown in Fig. 7. We, therefore, consider that the low-level features encoder and merged dilated convolutions can obtain better local- and global-context information and, thus, segment both multiscale objects (such as buildings and surface) and small classes (such as cars) accurately.

C. Generalization

Our models demonstrated very good generalization capabilities on both the Potsdam and Vaihingen data set. As shown in Table I, there are only −0.2% and −1.1% gaps in terms of mean F1-score between our local validation sets and the hold-out test sets of the Potsdam and Vaihingen, respectively. It is also worth noting that our model is the only one that works equally well on both Vaihingen IRRG data set and Potsdam RGB data set, which outperforms the DST_2 [21] model with 3.9% and 0.6% higher F1-score on Vaihingen and Potsdam data set, respectively, as shown in Tables II and III. Furthermore, our model achieves better performance (+0.5%) in terms of mean F1-score with fewer labeled training data than FuseNet+OSM [25] that used OpenStreetMap (OSM) as an additional data source.

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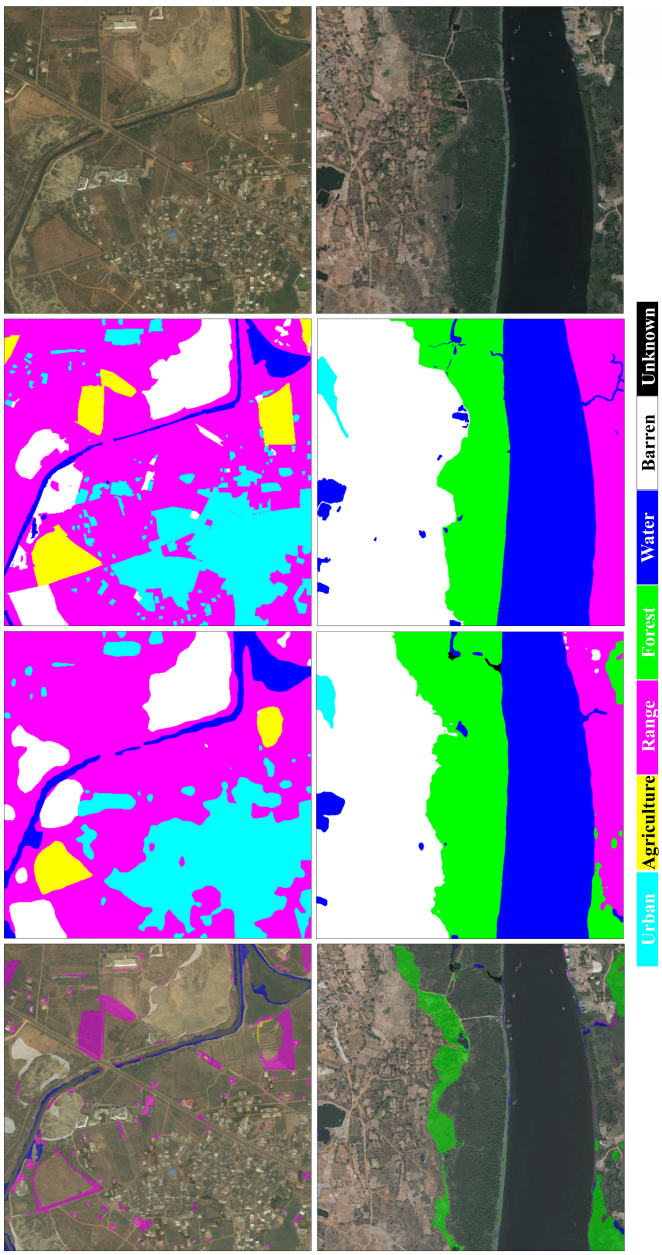


Fig. 8. Mapping results on local validation images of DeepGlobe. (From top to bottom) Input satellite images, the ground truths, the predictions of DDCM-SER50 model, and the errors of predictions.

However, we observed a bigger performance drop (approximately −15.9%) on the DeepGlobe data set when comparing results on the hold-out test set (mIoU: 56.2%) and local validation results (K-Fold avg. mIoU: 72.1%), as shown in Table VII. We see there is higher uncertainty (more false predictions) between range land (magenta), agriculture land (yellow), and forest (green) from the confusion matrix between classes for our model in Table VIII. Note that the model incorrectly classified some agricultural land to rangeland and predicted some rangeland as forest. These observations are also supported by the qualitative visualization of errors of our predictions, as shown in Fig. 8. Furthermore, in our experiments on the DeepGlobe data, we found that there are some annotation inaccuracies, mainly introduced by highly ambiguous objects and lower ground resolutions. What is worse is that we observed that the hold-out test images have different contrast and darker shadows than in the training set [30]. This obviously affected the model's final performance on the test sets.

VII. Conclusion

In this article, we presented a dense dilated convolutions' merging (DDCM) architecture for land cover classification for aerial imagery. The proposed architecture applies dilated convolutions to learn features at varying dilation rates and merges the feature map of each layer with the feature maps from all previous layers. On both the Potsdam and Vahingen data sets, our single model based on the DDCM-Net architecture achieves the best mean F1-score compared with the other architectures but with much fewer parameters and feature maps. DDCM-Net is easy to adapt to address a wide range of different problems, is fast to train, and achieves accurate results even on small data sets. The variants of our DDCM-Net by using different combinations of dilations and densities for the DeepGlobe data set also demonstrated better performance but consumed much fewer computation resources compared with other published methods.

Acknowledgment

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